

Threshold Entry in a Fixed-Prize Contest

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Manuscript skeleton, 7 October 2026

1 Introduction

Table 1: Headline claims, each traced to the model rows it rests on.

ID	Headline	Model rows	Scope note
K1	Every pure-strategy equilibrium has the same number of entrants, k^* , and the k^* richest challengers entering is always one of them.	M5, M6, M8	The count is pinned, and who enters is not, unless the condition of M7 holds. The count needs a gain that depends on the number of entrants alone (M26), and it fails when ability falls with wealth (M27).
K2	The effect of a spread of wealth on how many enter has a sign set by one comparison. Entry weakly rises when the marginal entrant stands at or above the pivot, and weakly falls when it stands at or below.	M12, M13, M15	Proved for pivot-spreads, and more generally under the margin condition. Inside the band the signs of the change settle nothing (M14), and the mean cannot replace the pivot (M13). Strict change is shown only for linear spreads (M15).
K3	The sorting needs only that the fee's utility cost falls with wealth, which strict concavity implies and which is strictly weaker.	M23, M24	No condition on u''' is used. Wealth sorting into a tournament is in print (Lazear and Rosen), through a condition on u''' , so the mechanism is not claimed as new.
K4	Entry is capped by the prize for every number of challengers. An entrant at index m with cost κ has $(m+1)\kappa \leq V$.	M10	The accounting behind the cap is in print for identical agents (Fu, Jiao and Lu; Fu and Lu). It replaces the old mobility ceiling and is not claimed as new.
K5	A larger prize does not lower the number of entrants, and a larger fee does not raise it.	M11	Weak monotonicity only.
K6	When the score gap is single-peaked, the first entrant's gain can rise and then fall in the number of challengers, and it never falls and then rises. With a uniform base score it never falls, for some base scores it must eventually fall, and stochastic dominance alone does not sign it.	M16, M17, M18, M19, M20, M21, M22	The kernel and the message that no such prediction is universal are Ryvkin and Drugov's. Whether the rise-then-fall result is new is pending the novelty reading (pass 7).

2 Model

Table 2: Results of the model, each with its statement and the Lean theorems that check it.

ID	Claim	Statement	Lean	Proof
M1	Representation (P1).	$\Delta(m) = V \int \varphi dH_m = V \mathbb{E}[\varphi(M_m)]$, with $M_m \sim H_m$.	EntryContestP1P2.two_max_sum, EntryContestP1P2.p1_representation, EntryContestP1P2.Delta_eq_expectation	p. 9

ID	Claim	Statement	Lean	Proof
M2	Dominance (P2).	For $s \geq 0$ and $\mu \leq 1$, $F \leq G$ and $\Delta(m) \geq 0$. For $\mu < 1$ and $\Pr(s > 0) > 0$, $F < G$ on some $[q, u)$ with $q < u$.	EntryContestP1P2.p2_fosd, EntryContestP1P2.p2_strict_interval, EntryContestP1P2.Delta_nonneg_from_primitives	p. 10
M3	Step identity (P3).	For $m + 2 \leq Q$, $\Delta(m+1) - \Delta(m) = -\frac{V}{2} \int \varphi^2 dK_m$.	EntryContestStep.three_max_sum_real, EntryContestStep.step_identity, EntryContestModel.Delta_step	p. 10
M4	Monotonicity.	$\Delta(m+1) \leq \Delta(m)$ for $V \geq 0$, with $<$ when $V > 0$ and $\int \varphi^2 dK_m > 0$.	EntryContestModel.Delta_step_nonpos, EntryContestModel.DeltaUpTo_step, EntryContestEq.Delta_step_strict	p. 10
M5	Threshold entry (P5).	For $\kappa_0 \leq \dots \leq \kappa_{Q-1}$ and Δ non-increasing, $\kappa_j \leq \Delta(j)$ and $i \leq j$ give $\kappa_i \leq \Delta(i)$.	EntryContest.single_crossing, EntryContestModel.p5_threshold_from_primitives	p. 10
M6	Count invariance.	Every equilibrium S , with $\kappa_i \leq \Delta(S - 1)$ for $i \in S$ and $\Delta(S) < \kappa_j$ for $j \notin S$, has $ S = k^*$.	EntryContestEq.count_invariance_fin, EntryContestMM.exists_mem_ge, EntryContestMM.exists_not_mem_le	p. 11
M7	Identities pinned.	If $\Delta(k^* - 1) < \kappa_{k^*}$, the assortative set $\{0, \dots, k^* - 1\}$ is the only equilibrium.	EntryContest.members_below_kstar, EntryContestEq.assortative_unique	p. 11
M8	Assortative selection (N4).	$\{0, \dots, k^* - 1\}$ is an equilibrium; $\sum_{i < k} \kappa_i \leq \sum_{i \in S} \kappa_i$ for $ S = k$; with identical score laws, payoff sums of equal-size equilibria differ by their cost sums.	EntryContestEq.assortative_is_equilibrium, EntryContestEq.assortative_min_cost, EntryContestEq.prize_term_anonymous, EntryContestEq.payoff_gap_anonymous	p. 11
M9	Benchmark (P6).	With the incumbent investing and $s \geq 0$ atomless, $\Delta(m) = V/(m+2)$ at $\mu = 0$ for every Q , and $\Delta(m) \rightarrow V/(m+2)$ as $\mu \rightarrow 0$.	EntryContestP6P7.integral_iid, EntryContestP6P7.integral_cdf_pow, EntryContestP6P7.p6_evaluation, EntryContestP6P7.p6_at_zero, EntryContestP6P7.tendsto_cdf_investorLaw, EntryContestP6P7.tendsto_cdf_nonInvestorLaw, EntryContestP6P7.p6_limit	p. 11

ID	Claim	Statement	Lean	Proof
M10	Saturation (P7).	$\Delta(m) \leq V/(m+1)$ for every Q , and an entrant at index m has $(m+1)\kappa \leq V$.	EntryContestP6P7. integral_cdf_pow, EntryContestP6P7.p7_ Delta_le, EntryContestP6P7.p7_ cap,EntryContestP6P7. p7_count	p. 12
M11	Prize and fee (P8).	k^* is non-decreasing in V and, for u non-decreasing, non-increasing in c .	EntryContestEq.kstar_ mono,EntryContestEq. Delta_mono_prize, EntryContestMM.Delta_ scale,EntryContestEq. kappa_mono_fee	p. 12
M12	Margin condition.	For sorted profiles w and w' , $w_{k^*-1} \leq w'_{k^*-1}$ gives $k^* \leq k'^*$, and $w'_{k^*} \leq w_{k^*}$ gives $k'^* \leq k^*$.	EntryContestSpread. enters_down, EntryContestSpread. count_ge_of_enters, EntryContestSpread. count_le_of_fails, EntryContestSpread. margin_rise, EntryContestSpread. margin_fall	p. 12
M13	Pivot spreads (P9-gen, P9).	For a pivot-spread T about x_0 , k^* weakly rises when $w_{k^*-1} \geq x_0$ and weakly falls when $w_{k^*-1} \leq x_0$. The linear spread about the mean is one, and keeps the mean. A change that keeps the total and majorizes w can lower k^* with w_{k^*-1} above the mean.	EntryContestSpread. pivot_rise, EntryContestSpread. pivot_fall, EntryContestSpread. spread_isPivot, EntryContestMM.mm2a, EntryContestSpread. mps_lowers_count	p. 13
M14	The band.	With $d = w' - w$, the signs of d leave the direction of k^* open only when $d_{k^*-1} < 0 < d_{k^*}$. A pivot-spread never does so. Inside the band two changes that keep the total and majorize w , with the same signs at every rank, move k^* in opposite directions for every small $\mu > 0$.	EntryContestSpread. branch_of_not_band, EntryContestSpread. not_band_of_single_ crossing, EntryContestSpread. pivot_single_crossing, EntryContestSpread. mps_band_signs, EntryContestSpread. mps_witness_counts, EntryContestSpread. mps_band_both_ directions, EntryContestSpread. mps_band_both_ directions_positive	p. 13
M15	Strictness under a spread (P9).	If some entrant lies below x_0 at $\lambda = 1$ and $\kappa(w) \rightarrow \infty$ as $w \downarrow c$, $k^*(\lambda)$ is non-increasing in $\lambda \geq 1$ and $k^*(\lambda) < k^*(1)$ above some $\bar{\lambda}$.	EntryContestKappa. exit_below_pivot, EntryContestKappa. enters_downward, EntryContestKappa.p9_ strict, EntryContestKappa.p9_ monotone, EntryContestKappa.p9_ strict_model	p. 13

ID	Claim	Statement	Lean	Proof
M16	Competitor number (P-MU).	For $Q \geq 1$ and every C , $D(Q) = -V(\int W_Q dF - \int W_Q dG)$, so for $V > 0$, $D(Q) > 0$ exactly when $\int W_Q dF < \int W_Q dG$. The kernel $g^n(1-g)$ rises on $[0, \frac{n}{n+1}]$ and falls on $[\frac{n}{n+1}, 1]$.	EntryContestPMU. weight_eq, EntryContestPMU.pmu_ identity, EntryContestPMU.pmu_ sign_iff, EntryContestPMU. kernel_rises, EntryContestPMU. kernel_falls, EntryContestPMU. kernel_max	p. 14
M17	Rise then fall in Q .	If $dF - dG \leq 0$ on $(-\infty, x_0]$ and $dF - dG \geq 0$ on (x_0, ∞) , and $V \geq 0$, then $D(Q_2) \leq G(x_0)^{Q_2-Q_1} D(Q_1)$ for $1 \leq Q_1 \leq Q_2$, and $\min(\Delta_{Q_1}(0), \Delta_{Q_3}(0)) \leq \Delta_{Q_2}(0)$ for $1 \leq Q_1 \leq Q_2 \leq Q_3$.	EntryContestPMU. weight_shift, EntryContestPMU.pmu_ single_crossing, EntryContestPMU.pmu_ step_nonpos_persists, EntryContestPMU. quasiconcave_of_steps, EntryContestPMU.pmu_ quasiconcave	p. 14
M18	Uniform base score.	For r uniform on $[0, 1]$, $s \geq 0$, $0 < \mu \leq 1$, $V \geq 0$ and every C , the hypothesis of M17 holds at $x_0 = \mu$, and $D(Q) \geq 0$ for $Q \geq 1$, so $\Delta_Q(0)$ is non-decreasing in $Q \geq 1$.	EntryContestPMU. uniform_left, EntryContestPMU. uniform_right, EntryContestPMU.pmu_ quasiconcave_uniform, EntryContestRefute. pmu_step_nonneg_ uniform, EntryContestRefute. Delta_mono_uniform	p. 14
M19	A class with rise then fall.	For $s \in \{0, 1\}$ with $\Pr(s=0) = p$, $0 < \mu < 1$, $V \geq 0$, $\Pr(r \leq 0) = 0$ and $\Pr(r \in B) \leq \Pr(r \in B - c)$ whenever $c > 0$ and $B \subseteq (c, \infty)$, the hypothesis of M17 holds at $x_0 = 1 - \mu$, and $\Delta_Q(0)$ is quasi-concave in $Q \geq 1$ for every C . A density of r non-increasing on $(0, \infty)$ gives the shift condition, and the smaller of two uniform draws has one.	EntryContestShift. left_of_lift, EntryContestShift. right_of_lift, EntryContestShift. quasiconcave_of_ shiftMono, EntryContestShift. shiftMono_of_density, EntryContestShift. lawA_shiftMono, EntryContestShift. witness_quasiconcave	p. 15
M20	The fall, exhibited.	For r the smaller of two uniform draws, $\mu = \frac{3}{4}$, s as in M19 and $C = F$, $D(1)$ and $D(2)$ are quadratics in p (in the proof). For $V > 0$, $p = \frac{1}{2}$ gives $\Delta_1(0) < \Delta_2(0)$ and $\Delta_3(0) < \Delta_2(0)$, and $p = \frac{1}{4}$ gives $\Delta_2(0) < \Delta_1(0)$.	EntryContestFall. stepQ_one, EntryContestFall. stepQ_two, EntryContestFall.rise_ then_fall, EntryContestFall.fall_ from_one	p. 15

ID	Claim	Statement	Lean	Proof
M21	When the gain must fall.	With s and r as in M19, $t = 1 - \mu$ and G_t the law of $\mu r + t$, if $t < a < b$, $G(t) < G(a)$, $G(b) < 1$, $C(a) > 0$ and $G((a, b]) < G_t((a, b])$, then for $V > 0$ and $p < 1$ some Q_0 has $D(Q) < 0$ for every $Q \geq Q_0$. The base law of M20 meets this on $(\frac{1}{2}, \frac{5}{8}]$ for every $p < 1$, and a uniform base score never does.	EntryContestMustFall. must_fall_general, EntryContestMustFall. shift_dominates, EntryContestMustFall. stepQ_bern, EntryContestMustFall. must_fall, EntryContestMustFall. witness_must_fall, EntryContestMustFall. unif_shiftMono, EntryContestMustFall. uniform_never_gains	p. 15
M22	Dominance does not sign $D(Q)$.	With $C = F$, the pairs $(F, G) = (x, 1 - (1 - x)^2)$ and (x^2, x) on $[0, 1]$ both have $F \leq G$ and no atoms. The first has $D(1) = \frac{V}{60}$ and $D(2) = \frac{V}{420}$, and the second has $D(Q) = -\frac{VQ}{(Q+2)(Q+3)(Q+4)}$.	EntryContestPMUWitness. pairA_fosd, EntryContestPMUWitness. pairB_fosd, EntryContestPMUWitness. pairA_step_one, EntryContestPMUWitness. pairA_step_two, EntryContestPMUWitness. pairB_step, EntryContestPMUWitness. fosd_does_not_sign	p. 16
M23	Burden-monotonicity.	$\kappa(w) = u(w) - u(w - c)$ is strictly decreasing on $w > c$ when u is strictly concave on $(0, \infty)$, or when $u'(w) < u'(w - c)$ for every $w > c$. A convex stretch of u longer than c rules it out. Log and CRRA utility ($\gamma > 0$, $\gamma \neq 1$) satisfy it.	EntryContestKappa. burden_strictAnti_of_ strictConcave, EntryContestBM.burden_ strictAnti_of_deriv, EntryContestBM.not_ burden_of_convex_ stretch, EntryContestKappa.log_ burden_strictAnti, EntryContestKappa. crra_strictConcave, EntryContestKappa. crra_burden_strictAnti	p. 16
M24	Weaker than concavity.	For every $c > 0$ some strictly increasing u , not concave on $(0, \infty)$, has κ strictly decreasing on (c, ∞) and $\kappa(w) \rightarrow \infty$ as $w \downarrow c$. The parameters of the construction form an open set, and its convex stretch can have any length below c .	EntryContestBM.rampU_ burden, EntryContestBM. rampU_kappa_diverges, EntryContestBM.rampU_ strictConvex, EntryContestBM.rampU_ not_concave, EntryContestBM. admissibleParams_open, EntryContestBM.rampU_ admissible, EntryContestBM. admissible_width, EntryContestBM.bm_ strictly_weaker	p. 16

ID	Claim	Statement	Lean	Proof
M25	Divergence at the fee.	For u continuous at c , $\kappa(w) \rightarrow \infty$ as $w \downarrow c$ exactly when $u(x) \rightarrow -\infty$ as $x \downarrow 0$. Log utility and CRRA with $\gamma > 1$ satisfy it, and CRRA with $\gamma < 1$ does not.	EntryContestKappa.kappa_diverges_iff, EntryContestKappa.not_diverges_of_continuousWithinAt_zero, EntryContestKappa.log_kappa_diverges, EntryContestKappa.crra_kappa_diverges, EntryContestKappa.crra_kappa_not_diverges	p. 17
M26	Anonymity.	If every entrant's score has the law F , challenger i 's gain from entering against the entrants T is $\Delta(T)$, whoever they are.	EntryContestAnon.profile_mem, EntryContestAnon.profile_not_mem, EntryContestAnon.cdf_profile, EntryContestAnon.gain_anonymous	p. 17
M27	The anonymity boundary.	If the law of s_j varies with j , the gain depends on who the other entrants are. With ability falling in wealth, an equilibrium of one entrant and an equilibrium of two coexist for every small $\mu > 0$.	EntryContestAnon.winProb_zero_entrant, EntryContestAnon.winProb_zero_outsider, EntryContestAnon.gain_zero, EntryContestAnon.kappa3_values, EntryContestAnon.gains3, EntryContestAnon.anon_witness_zero, EntryContestAnon.gain_tendsto, EntryContestAnon.anon_witness_positive, EntryContestAnon.anon_witness_exists	p. 17

3 Literature

Table 3: Papers the model or the conclusions depend on, each with a verbatim quote.

ID	Paper	What we rely on	Quote	Page	Model rows
L1	Lazear and Rosen [1981]	Wealth sorting into a tournament is in print, by example and through declining absolute risk aversion, a condition on u''' . M23 sorts through κ alone, which $u'' < 0$ gives.	"Individuals will self-select the payment scheme in accordance with their wealth."	855	M5, M23
L2	Levin and Smith [1994]	With identical potential entrants, pure strategies leave who enters unexplained, and mixed entry makes the count random.	"The process by which potential bidders divide into these two groups is not explained"	586	M6, M7

ID	Paper	What we rely on	Quote	Page	Model rows
L3	Morgan et al. [2012]	With identical players the entrants' identities are not pinned, and sequential entry orders them.	"the identity of the players choosing to opt into the contest is not uniquely determined"	441	M6, M7
L4	Moreno and Wooders [2011]	Under private entry costs every buyer compares her cost with one threshold, and the count is binomial. Here the threshold depends on the index.	"indicating the maximum entry cost for which the buyer enters the auction"	319–320	M5, M6
L5	Fullerton and McAfee [1999]	With heterogeneous types the efficient entry equilibrium admits the lowest-cost firms. Profits depend on rivals' types, and entry equilibria of sizes 2 and 3 coexist (FullertonMcAfee.two_sizes), so count invariance needs anonymity.	"entry of the m lowest-cost firms is an equilibrium"	580	M6, M7, M26
L6	Fu et al. [2015]	The rent-accounting bound behind M10, for identical agents with mixed entry, capping an expected count.	"the rent to be dissipated in the contest cannot exceed"	397	M10
L7	Fu and Lu [2010]	Total effort equals the budget less the entrants' total entry cost, in a contest chosen by a designer. <i>Unverified</i> : the working paper (MPRA 945) was read, and the <i>Economic Inquiry</i> version is not checked.	"the equilibrium total effort is given by the difference between the total budget of the contest organizer and the total entry costs incurred by participating contestants, regardless of the contest technology."	10	M10
L8	Ryvkin and Drugov [2020]	The order-statistic kernel, and that no prediction in the number of players holds for every noise law. M16 to M22 are the entry-gain instance.	"There is not a single prediction, either for individual or aggregate effort, that cannot be reversed for at least some distribution of noise."	1589	M16, M17, M22
L9	Costrell and Loury [2004]	A spread's effect signed by the position of one marginal quantile, which is fixed by technology there and endogenous in M12. <i>Unverified</i> : a draft was read, and the published version is not checked.	"completely characterize the technology"	6	M12, M13
L10	Hopkins and Kornienko [2010]	The dispersive order as the order for inequality comparisons, quoting Shaked and Shanthikumar (2007).	"A variable with distribution F is said to be smaller in the dispersive order (or less dispersed) than a variable with a distribution G "	128	M13, M14
L11	Lewis and Thompson [1981]	The same order, defined earlier by quantile spacing, which with a crossing point gives the pivot-spreads of M13.	"We call the relationship defined by (1.1), (1.2) an ordering in dispersion of G and H "	77	M13, M14

ID	Paper	What we rely on	Quote	Page	Model rows
L12	Schroyen and Treich [2016]	The privilege-contest payoff, with a prize valued apart from wealth, and an intensive-margin effect of a spread that turns on u''' . <i>Unverified</i> : the working paper (No. 16-699) was read, and the <i>Games and Economic Behavior</i> version is not checked.	“and therefore its marginal value is independent of the level of wealth”	8	M13, M23
L13	Cole et al. [1992]	A more compact income distribution raises savings through matching on wealth. The direction matches the below-pivot branch of M13, and the mechanism differs.	“the agent from the economy with the more compact income distribution will tend to have a higher savings rate”	1103	M13

Papers cited in our other documents and not read are not in the table, and every claim made about them is unverified. They are Shaked [1982], which Hopkins and Kornienko cite for the dispersive order, Mathews and Namoro [2008], Corcoran [1984] and Corcoran and Karels [1985].

4 Conclusions

The conclusions come in two tables. Table 4 holds the claims that rest on the model rows and on papers read in full. Table 5 holds the claims that something is new. Each of those waits on a paper not yet read in full, or on a comparison not yet done, and says which. The novelty reading is pass 7 of `TODO.md`.

Table 4: Conclusions that rest on the model rows and on papers read in full.

ID	Statement	Rests on
C1	For a fixed prize awarded by rank, a spread of wealth changes the number of entrants in a direction set by one comparison, the marginal entrant against the pivot of the spread, and the rule needs only that the fee’s utility cost falls with wealth. In the privilege contest of Schroyen and Treich the effect of a spread on effort turns on u''' , so the entry margin is what frees the sign from the third derivative. Their paper was read in its working-paper version (L12).	M12, M13, M23, M24, L12
C2	The rule reads the wealth profile at two ranks only, the marginal entrant and the first outsider, and it holds for any change of profile, mean-preserving or not. Where the marginal entrant becomes poorer and the first outsider richer, the signs of the change settle nothing, even for mean-preserving spreads, and the mean cannot stand in for the pivot. Costrell and Loury sign a spread by one marginal quantile fixed by technology, and here the margin is the count itself, which moves with the profile. Their paper was read as a draft (L9).	M12, M13, M14, L9
C3	Every pure-strategy equilibrium has the same number of entrants because the gain depends on that number alone. When the gain depends on who enters, the count is not pinned. This happens in the model when ability falls with wealth, and in the entry stage of Fullerton and McAfee, where profits depend on rivals’ costs.	M6, M26, M27, L5
C4	Under complete information, heterogeneous entry costs pin the count without the devices that models with identical entrants use (mixed entry, sequential arrival, private random delays), and the k^* richest entering is always an equilibrium. Who enters is pinned only under the further condition of M7.	M5, M6, M7, M8, L2, L3, L4

Table 5: Claims pending verification. Each says what it waits on.

ID	Statement	Rests on
C5	The sign rule of C1 is new. None of the entry papers read signs the effect of a spread of wealth, or of entry costs, on the number of entrants at a fixed number of potential entrants. <i>Pending</i> : Mathews and Namoro [2008], Corcoran [1984] and Corcoran and Karels [1985], which are entry papers and are not read.	M13, L2, L3, L4, L5, L6, L7
C6	Count invariance across pure-strategy equilibria with heterogeneous entry costs is new. The entry papers read have identical entrants (Levin and Smith, Morgan, Orzen and Sefton, Fu, Jiao and Lu, Fu and Lu), a random count under private costs (Moreno and Wooders), or, in Fullerton and McAfee, a count that differs across equilibria. <i>Pending</i> : the same three papers.	M6, L2, L3, L4, L5, L6, L7
C7	That the first entrant's gain has no interior minimum in the number of challengers when the score gap is single-peaked, never falls with a uniform base score, and must eventually fall for some base scores, is new as a statement about the gain from entry. The kernel and the message that no such prediction is universal are Ryvkin and Drugov's. <i>Pending</i> : a comparison with their unimodality result for individual effort, which may contain this one as an instance.	M16, M17, M18, M19, M20, M21, L8
C8	That two ranks suffice to sign the change in the count is a sharpening that the assignment literature does not have. <i>Pending</i> : the published version of Costrell and Loury, read here as a draft.	M12, L9

A Proofs

Notation. Challenger i scores $\sigma_i = \mu r_i + (1 - \mu)s_i e_i$, with r_i and $s_i \geq 0$ drawn independently of wealth. F and G are the score CDFs of an entrant and of an outsider, and C is the incumbent's, each without atoms. With m other entrants among the Q challengers,

$$H_m = C F^m G^{Q-1-m}, \quad K_m = C F^m G^{Q-2-m}, \quad \varphi = G - F,$$

$$\Delta(m) = V \left(\int H_m dF - \int H_m dG \right), \quad \kappa(w) = u(w) - u(w - c).$$

H_m is the CDF of the best score facing a challenger, M_m a random variable with that CDF, and K_m the same CDF with one more challenger set aside, so that $H_{m+1} = K_m F$ and $H_m = K_m G$. Challengers are indexed $0, \dots, Q-1$ in increasing order of cost, $\kappa_0 \leq \dots \leq \kappa_{Q-1}$, which is decreasing order of wealth. Index j enters when $\kappa_j \leq \Delta(j)$, and k^* satisfies $\kappa_j \leq \Delta(j)$ for $j < k^*$ and $\Delta(k^*) < \kappa_{k^*}$ when $k^* < Q$. In the results on wealth profiles, rank j has wealth w_j with $w_0 \geq w_1 \geq \dots$, and enters when $w_j > c$ and $\kappa(w_j) \leq \Delta(j)$, so a challenger whose wealth is at or below the fee does not enter. A prime marks the profile after a change and its count k'^* . When the number of challengers varies, $\Delta_Q(0)$ is the first entrant's gain among Q challengers, with

$$D(Q) = \Delta_{Q+1}(0) - \Delta_Q(0), \quad W_Q = C G^{Q-1} (1 - G).$$

Proof of M1

Lean: EntryContestP1P2.two_max_sum, EntryContestP1P2.p1_representation, EntryContestP1P2.Delta_eq_expectation.

For independent $X \sim A$ and $Y \sim B$ with B atomless, $\Pr(X = Y) = 0$, so

$$\int B dA + \int A dB = \Pr(Y \leq X) + \Pr(X \leq Y) = 1.$$

With $(A, B) = (F, H_m)$ and $(A, B) = (G, H_m)$,

$$\int H_m dF = 1 - \int F dH_m, \quad \int H_m dG = 1 - \int G dH_m,$$

so

$$\frac{\Delta(m)}{V} = \int (G - F) dH_m = \int \varphi dH_m = \mathbb{E}[\varphi(M_m)].$$

□

Proof of M2

Lean: EntryContestP1P2.p2_fosd, EntryContestP1P2.p2_strict_interval, EntryContestP1P2.Delta_nonneg_from_primitives.

Let $X = \mu r + (1 - \mu)s$ and $Y = \mu r$. For $\mu \leq 1$ and $s \geq 0$, $X - Y = (1 - \mu)s \geq 0$, so $\{X \leq x\} \subseteq \{Y \leq x\} \cup \{s < 0\}$ and

$$F(x) \leq G(x) + \Pr(s < 0) = G(x), \quad \Delta(m) = V \int \phi dH_m \geq 0.$$

For $\mu < 1$,

$$\{s > 0\} \subseteq \{Y < X\} = \bigcup_{q \in \mathbb{Q}} \{Y \leq q < X\},$$

so $\Pr(s > 0) > 0$ gives $q \in \mathbb{Q}$ with $\Pr(Y \leq q < X) > 0$, and

$$G(q) \geq \Pr(X \leq q) + \Pr(Y \leq q < X) > F(q).$$

$G - F$ is right-continuous, so $G - F > 0$ on some $[q, u)$. □

Proof of M3

Lean: EntryContestStep.three_max_sum_real, EntryContestStep.step_identity, EntryContestModel.Delta_step.

Write $K = K_m$, so $H_{m+1} = KF$ and $H_m = KG$. For independent atomless draws $Z \sim K$, $X \sim A$, $Y \sim B$, exactly one is the largest,

$$\int AB dK + \int KB dA + \int KA dB = 1.$$

With $(A, B) = (F, F)$, (G, G) and (F, G) ,

$$\int KF dF = \frac{1}{2} \left(1 - \int F^2 dK \right), \quad \int KG dG = \frac{1}{2} \left(1 - \int G^2 dK \right), \quad \int KG dF + \int KF dG = 1 - \int FG dK,$$

so

$$\frac{\Delta(m+1) - \Delta(m)}{V} = \int KF dF - \int KF dG - \int KG dF + \int KG dG = -\frac{1}{2} \int (F - G)^2 dK. \quad \square$$

Proof of M4

Lean: EntryContestModel.Delta_step_nonpos, EntryContestModel.DeltaUpTo_step, EntryContestEq.Delta_step_strict.

By M3 and $\phi^2 \geq 0$,

$$\Delta(m+1) - \Delta(m) = -\frac{V}{2} \int \phi^2 dK_m \leq 0 \quad (V \geq 0), \quad < 0 \quad \left(V > 0, \int \phi^2 dK_m > 0 \right).$$

Beyond $m = Q - 1$ the gain is set to $\Delta(Q - 1)$, so $\Delta(m+1) \leq \Delta(m)$ for every m . □

Proof of M5

Lean: EntryContest.single_crossing, EntryContestModel.p5_threshold_from_primitives.

For $i \leq j$, by M4,

$$\kappa_i \leq \kappa_j \leq \Delta(j) \leq \Delta(i). \quad \square$$

Proof of M6

Lean: EntryContestEq.count_invariance_fin, EntryContestMM.exists_mem_ge, EntryContestMM.exists_not_mem_le.

Let S be an equilibrium and $k = |S|$. A k -set of indices has an element $i \geq k - 1$, since $S \subseteq \{0, \dots, k - 2\}$ gives $k \leq k - 1$, and omits some $j \leq k$, since $\{0, \dots, k\} \subseteq S$ gives $k + 1 \leq k$. If $k > k^*$, then $k^* < Q$ and

$$\Delta(k^*) < \kappa_{k^*} \leq \kappa_{k-1} \leq \kappa_i \leq \Delta(k - 1) \leq \Delta(k^*).$$

If $k < k^*$, then $j < Q$ and

$$\Delta(k) < \kappa_j \leq \kappa_k \leq \Delta(k).$$

Both are contradictions, so $k = k^*$. □

Proof of M7

Lean: EntryContest.members_below_kstar, EntryContestEq.assortative_unique.

A member i of an equilibrium has $\kappa_i \leq \Delta(k^* - 1)$ by M6. If $i \geq k^*$,

$$\kappa_i \geq \kappa_{k^*} > \Delta(k^* - 1),$$

so $i < k^*$. Then $S \subseteq \{0, \dots, k^* - 1\}$ and $|S| = k^*$ give $S = \{0, \dots, k^* - 1\}$. □

Proof of M8

Lean: EntryContestEq.assortative_is_equilibrium, EntryContestEq.assortative_min_cost, EntryContestEq.prize_term_anonymous, EntryContestEq.payoff_gap_anonymous.

(i) For $i < k^*$ and for $k^* \leq j < Q$,

$$\kappa_i \leq \kappa_{k^*-1} \leq \Delta(k^* - 1), \quad \Delta(k^*) < \kappa_{k^*} \leq \kappa_j.$$

(ii) Induct on S , adding its largest element a last. With $T = S \setminus \{a\}$ and $T \subseteq \{0, \dots, a - 1\}$, $|T| \leq a$, so

$$\sum_{i < |T|+1} \kappa_i = \sum_{i < |T|} \kappa_i + \kappa_{|T|} \leq \sum_{i \in T} \kappa_i + \kappa_a = \sum_{i \in S} \kappa_i.$$

(iii) With identical laws, the win probability of i is $P^{\text{in}}(|S|) = \int CF^{|S|-1} G^{Q-|S|} dF$ for $i \in S$ and $P^{\text{out}}(|S|) = \int CF^{|S|} G^{Q-1-|S|} dG$ for $i \notin S$, so with $EU_i(S) = u(w_i) - \kappa_i \mathbf{1}\{i \in S\} + VP_i(S)$,

$$\sum_i EU_i(S) = \sum_i u(w_i) - \sum_{i \in S} \kappa_i + V(|S|P^{\text{in}}(|S|) + (Q - |S|)P^{\text{out}}(|S|)),$$

and the last term is the same for any two entrant sets of one size. □

Proof of M9

Lean: EntryContestP6P7.integral_iid, EntryContestP6P7.integral_cdf_pow, EntryContestP6P7.p6_evaluation, EntryContestP6P7.p6_at_zero, EntryContestP6P7.tendsto_cdf_investorLaw, EntryContestP6P7.tendsto_cdf_nonInvestorLaw, EntryContestP6P7.p6_limit.

For $X_0, \dots, X_k$ independent with atomless CDF F ,

$$\int F^k dF = \Pr\left(X_0 = \max_{i \leq k} X_i\right) = \frac{1}{k+1}.$$

At $\mu = 0$ the entrant scores s and the outsider 0, so $G = \mathbf{1}_{[0,\infty)}$ and $F(0) = \Pr(s \leq 0) = 0$. With $C = F$,

$$\int H_m dF = \int F^{m+1} dF = \frac{1}{m+2}, \quad \int H_m dG = H_m(0) = 0.$$

For $\mu \rightarrow 0$ let $X_\mu = \mu r + (1 - \mu)s$ and $Y_\mu = \mu r$. For $s > 0$,

$$F_\mu(X_\mu) \rightarrow F_0(s), \quad G_\mu(X_\mu) \rightarrow 1, \quad F_\mu(Y_\mu) \rightarrow F_0(0) = 0,$$

with every integrand in $[0, 1]$, so by dominated convergence

$$\int H_m dF_\mu \rightarrow \int F_0^{m+1} dF_0 = \frac{1}{m+2}, \quad 0 \leq \int H_m dG_\mu \leq \mathbb{E}[F_\mu(Y_\mu)] \rightarrow 0.$$

□

Proof of M10

Lean: EntryContestP6P7.integral_cdf_pow, EntryContestP6P7.p7_Delta_le, EntryContestP6P7.p7_cap, EntryContestP6P7.p7_count.

From $C \leq 1$, $G \leq 1$ and $\int H_m dG \geq 0$,

$$\Delta(m) \leq V \int H_m dF \leq V \int F^m dF = \frac{V}{m+1}.$$

An entrant at index m with cost κ has $\kappa \leq \Delta(m)$, so

$$(m+1)\kappa \leq (m+1)\Delta(m) \leq V, \quad m+1 \leq \frac{V}{\kappa} \quad (\kappa > 0).$$

□

Proof of M11

Lean: EntryContestEq.kstar_mono, EntryContestEq.Delta_mono_prize, EntryContestMM.Delta_scale, EntryContestEq.kappa_mono_fee.

Let k^* and k'^* be the counts for (κ, Δ) and (κ', Δ') , with $\kappa'_j \leq \kappa_j$ and $\Delta(j) \leq \Delta'(j)$ for every j . If $k'^* < k^*$, then $k'^* < Q$ and

$$\Delta'(k'^*) < \kappa'_{k'^*} \leq \kappa_{k'^*} \leq \Delta(k'^*) \leq \Delta'(k'^*),$$

so $k^* \leq k'^*$. Write Δ_V for the gain at prize V . By M1 and M2, $\Delta_V(j) = V \Delta_1(j)$ with $\Delta_1(j) \geq 0$, and for u non-decreasing,

$$V \leq V' \Rightarrow \Delta_V \leq \Delta_{V'}, \quad c \leq c' \Rightarrow u(w) - u(w - c) \leq u(w) - u(w - c').$$

□

Proof of M12

Lean:

EntryContestSpread.enters_down, EntryContestSpread.count_ge_of_enters, EntryContestSpread.count_le_of_fails, EntryContestSpread.margin_rise, EntryContestSpread.margin_fall.

(i) The marginal entrant has $c < w_{k^*-1} \leq w'_{k^*-1}$, so

$$\kappa(w'_{k^*-1}) \leq \kappa(w_{k^*-1}) \leq \Delta(k^* - 1),$$

and for $i \leq k^* - 1$, with $w'_i \geq w'_{k^*-1} > c$,

$$\kappa(w'_i) \leq \kappa(w'_{k^*-1}) \leq \Delta(k^* - 1) \leq \Delta(i).$$

Every $i < k^*$ enters after the change, so $k^* \leq k'^*$. (ii) For $k^* < Q$, rank k^* fails before the change. If $w'_{k^*} > c$, then $c < w'_{k^*} \leq w_{k^*}$ and

$$\kappa(w'_{k^*}) \geq \kappa(w_{k^*}) > \Delta(k^*),$$

so rank k^* fails after the change and $k'^* \leq k^*$.

□

Proof of M13

Lean: EntryContestSpread.pivot_rise, EntryContestSpread.pivot_fall, EntryContestSpread.spread_isPivot, EntryContestMM.mm2a, EntryContestSpread.mps_lowers_count.

T is non-decreasing, so $w'_j = T(w_j)$ is sorted, and

$$x_0 \leq w_{k^*-1} \Rightarrow w_{k^*-1} \leq T(w_{k^*-1}), \quad w_{k^*} \leq w_{k^*-1} \leq x_0 \Rightarrow T(w_{k^*}) \leq w_{k^*},$$

so M12 applies. For $T(w) = x_0 + \lambda(w - x_0)$ with $\lambda \geq 1$,

$$T(w) - w = (\lambda - 1)(w - x_0), \quad \frac{1}{Q} \sum_j T(w_j) = \bar{w} + \lambda \left(\frac{1}{Q} \sum_j w_j - \bar{w} \right) = \bar{w} \quad (x_0 = \bar{w}).$$

With $c = 1$, $\kappa(w) = \frac{1}{w-1} - \frac{1}{w}$ and $\Delta(m) = \frac{1}{m+2}$,

$$w = \left(\frac{29}{10}, \frac{14}{5}, 2\right), \quad w' = \left(\frac{18}{5}, \frac{11}{5}, \frac{19}{10}\right), \quad \sum_j w_j = \sum_j w'_j, \quad w'_2 \leq w_2, \quad w'_2 + w'_1 \leq w_2 + w_1,$$

$$\kappa(w) = \left(\frac{100}{551}, \frac{25}{126}, \frac{1}{2}\right), \quad \kappa(w') = \left(\frac{25}{234}, \frac{25}{66}, \frac{100}{171}\right), \quad \frac{1}{3} \sum_j w_j = \frac{77}{30} < w_1,$$

so the counts are 2 and 1. □

Proof of M14

Lean: EntryContestSpread.branch_of_not_band, EntryContestSpread.not_band_of_single_crossing, EntryContestSpread.pivot_single_crossing, EntryContestSpread.mps_band_signs, EntryContestSpread.mps_witness_counts, EntryContestSpread.mps_band_both_directions, EntryContestSpread.mps_band_both_directions_positive.

Outside the band $d_{k^*-1} \geq 0$ or $d_{k^*} \leq 0$, and M12 signs the change. If

$$i \leq j, d_j > 0 \Rightarrow d_i \geq 0,$$

then $d_{k^*} > 0$ gives $d_{k^*-1} \geq 0$. A pivot-spread has this property, since $d_j > 0$ gives $w_j > x_0$ and then $w_i \geq w_j > x_0$. With κ and Δ as in M13, the profiles

$$w = \left(4, \frac{14}{5}, \frac{11}{5}, 2\right), \quad w' = \left(\frac{41}{10}, \frac{27}{10}, \frac{13}{5}, \frac{8}{5}\right), \quad w'' = \left(\frac{23}{5}, \frac{9}{4}, \frac{9}{4}, \frac{19}{10}\right)$$

all sum to 11, with partial sums from the poorest up

$$\left(2, \frac{21}{5}, 7\right), \quad \left(\frac{8}{5}, \frac{21}{5}, \frac{69}{10}\right), \quad \left(\frac{19}{10}, \frac{83}{20}, \frac{32}{5}\right),$$

and

$$\kappa(w) = \left(\frac{1}{12}, \frac{25}{126}, \frac{25}{66}, \frac{1}{2}\right), \quad \kappa(w') = \left(\frac{100}{1271}, \frac{100}{459}, \frac{25}{104}, \frac{25}{24}\right), \quad \kappa(w'') = \left(\frac{25}{414}, \frac{16}{45}, \frac{16}{45}, \frac{100}{171}\right)$$

against $\Delta = \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}\right)$. The counts are 2, 3 and 1, and d' and d'' both have signs $(+, -, +, -)$. Every inequality is strict, so by M9 the counts hold for every small $\mu > 0$. □

Proof of M15

Lean:

EntryContestKappa.exit_below_pivot, EntryContestKappa.enters_downward, EntryContestKappa.p9_strict, EntryContestKappa.p9_monotone, EntryContestKappa.p9_strict_model.

Let j be an entrant at $\lambda = 1$ with $w_j < x_0$. Since $\kappa(v) \rightarrow \infty$ as $v \downarrow c$, some $u > c$ has $\kappa > \Delta(j)$ on (c, u) , and

$$\lambda > \bar{\lambda} = \max \left(1, \frac{x_0 - u}{x_0 - w_j}\right) \Rightarrow x_0 + \lambda(w_j - x_0) < u,$$

so rank j fails at λ . By the prefix property no rank from j on enters, and

$$k^*(\lambda) \leq j < j+1 \leq k^*(1).$$

For $1 \leq \lambda_1 \leq \lambda_2$, ranks with $w_i \geq x_0$ have $i < j$ and enter at λ_1 , while ranks with $w_i < x_0$ have $x_0 + \lambda_2(w_i - x_0) \leq x_0 + \lambda_1(w_i - x_0)$, so every entrant at λ_2 enters at λ_1 . □

Proof of M16

Lean:

EntryContestPMU.weight_eq, EntryContestPMU.pmu_identity, EntryContestPMU.pmu_sign_iff, EntryContestPMU.kernel_rises, EntryContestPMU.kernel_falls, EntryContestPMU.kernel_max.

With no other entrant, $H_0 = CG^{Q-1}$ among Q challengers and CG^Q among $Q+1$, and $CG^{Q-1} - CG^Q = W_Q$, so

$$\frac{D(Q)}{V} = \int (CG^Q - CG^{Q-1}) dF - \int (CG^Q - CG^{Q-1}) dG = - \left(\int W_Q dF - \int W_Q dG \right).$$

For the kernel,

$$\frac{d}{dg} g^n (1-g) = g^{n-1} (n - (n+1)g) \quad (n \geq 1), \quad \frac{d}{dg} (1-g) = -1 \quad (n=0).$$

□

Proof of M17

Lean: EntryContestPMU.weight_shift, EntryContestPMU.pmu_single_crossing, EntryContestPMU.pmu_step_nonpos_persists, EntryContestPMU.quasiconcave_of_steps, EntryContestPMU.pmu_quasiconcave.

Let $k = Q_2 - Q_1$, $\rho = G(x_0)^k$ and $v = dF - dG$. By M16 and $W_{Q_2} = G^k W_{Q_1}$,

$$D(Q_2) - \rho D(Q_1) = -V \int (G^k - \rho) W_{Q_1} dv.$$

G is non-decreasing and $W_{Q_1} \geq 0$, so

$$(G^k - \rho) W_{Q_1} \leq 0, \quad v \leq 0 \text{ on } (-\infty, x_0], \quad (G^k - \rho) W_{Q_1} \geq 0, \quad v \geq 0 \text{ on } (x_0, \infty),$$

and $D(Q_2) \leq \rho D(Q_1)$. Then $D(i) \leq 0$ gives $D(j) \leq 0$ for $j \geq i$. If $D(i) \leq 0$ for some $Q_1 \leq i < Q_2$, then $\Delta_{Q_3}(0) \leq \Delta_{Q_2}(0)$, and otherwise $\Delta_{Q_1}(0) \leq \Delta_{Q_2}(0)$.

□

Proof of M18

Lean: EntryContestPMU.uniform_left, EntryContestPMU.uniform_right, EntryContestPMU.pmu_quasiconcave_uniform, EntryContestRefute.pmu_step_nonneg_uniform, EntryContestRefute.Delta_mono_uniform.

G is the law of μr , so $G = 1$ on $[\mu, \infty)$, $dG = 0$ on (μ, ∞) and $W_Q = 0$ there. For $B \subseteq (-\infty, \mu]$ and $y \geq 0$, with $\tau = (1 - \mu)y/\mu$ and λ Lebesgue measure,

$$\Pr(\mu r + (1 - \mu)y \in B) = \lambda\left([\tau, 1] \cap \frac{B}{\mu}\right) \leq \lambda\left([0, 1] \cap \frac{B}{\mu}\right) = \Pr(\mu r \in B),$$

so $dF \leq dG$ on $(-\infty, \mu]$ and M17 applies at $x_0 = \mu$. By M16,

$$\frac{D(Q)}{V} = \int_{(-\infty, \mu]} W_Q d(G - F) \geq 0.$$

□

Proof of M19

Lean: EntryContestShift.left_of_lift, EntryContestShift.right_of_lift, EntryContestShift.quasiconcave_of_shiftMono, EntryContestShift.shiftMono_of_density, EntryContestShift.lawA_shiftMono, EntryContestShift.witness_quasiconcave.

Let $t = 1 - \mu$ and G_t be the law of $\mu r + t$, so $F = pG + (1 - p)G_t$. From $\Pr(r \leq 0) = 0$, G_t has no mass on $(-\infty, t]$, where $dF = p dG \leq dG$. For $B \subseteq (t, \infty)$, with $c = t/\mu$ and $B/\mu \subseteq (c, \infty)$,

$$G(B) = \Pr(r \in \frac{B}{\mu}) \leq \Pr(r \in \frac{B}{\mu} - c) = G_t(B),$$

so $dF \geq dG$ on (t, ∞) and M17 applies at $x_0 = t$. For a density g non-increasing on $(0, \infty)$ and $B \subseteq (c, \infty)$,

$$\int_B g(x) dx \leq \int_B g(x - c) dx = \int_{B-c} g(x) dx,$$

and the smaller of two uniform draws has $\Pr(r \leq x) = 1 - (1 - x)^2$ and density $2(1 - x)$ on $[0, 1]$. □

Proof of M20

Lean: EntryContestFall.stepQ_one, EntryContestFall.stepQ_two, EntryContestFall.rise_then_fall, EntryContestFall.fall_from_one.

Here $G(x) = 1 - (1 - \frac{4x}{3})^2$ on $[0, \frac{3}{4}]$, $F(x) = pG(x) + (1 - p)G(x - \frac{1}{4})$, $W_1 = F(1 - G)$ and $W_2 = FG(1 - G)$. By M16, integrating the piecewise polynomials on $[0, \frac{1}{4}]$, $[\frac{1}{4}, \frac{3}{4}]$ and $[\frac{3}{4}, 1]$,

$$\frac{D(1)}{V} = -\frac{11p^2}{162} + \frac{1901p}{21870} - \frac{208}{10935}, \quad \frac{D(2)}{V} = -\frac{4933p^2}{393660} + \frac{77219p}{2755620} - \frac{10672}{688905},$$

so

$$p = \frac{1}{2} : \frac{D(1)}{V} = \frac{653}{87480}, \quad \frac{D(2)}{V} = -\frac{10169}{2204496}, \quad p = \frac{1}{4} : \frac{D(1)}{V} = -\frac{179}{116640}.$$

□

Proof of M21

Lean: EntryContestMustFall.must_fall_general, EntryContestMustFall.shift_dominates, EntryContestMustFall.stepQ_bern, EntryContestMustFall.must_fall, EntryContestMustFall.witness_must_fall, EntryContestMustFall.unif_shiftMono, EntryContestMustFall.uniform_never_gains.

With $F = pG + (1 - p)G_t$ and $W_Q = G^{Q-1}W_1$, M16 gives

$$D(Q) = -V(1 - p) \left(\int W_Q dG_t - \int W_Q dG \right).$$

G_t has no mass on $(-\infty, t]$, $dG_t \geq dG$ on (t, ∞) by M19, and $W_1 \geq \eta = C(a)(1 - G(b)) > 0$ on $(a, b]$. Splitting at t and a ,

$$\int W_Q dG_t - \int W_Q dG \geq G(a)^{Q-1}A - G(t)^{Q-1}B,$$

$$A = \int_{(a, \infty)} W_1 d(G_t - G) \geq \eta (G_t - G)((a, b]) > 0, \quad B = \int_{(-\infty, t]} W_1 dG \geq 0.$$

With $r_0 = G(t)/G(a) < 1$ and $r_0^{Q_0-1}(B+1) < A$, the bound is positive for $Q \geq Q_0$. For M20's base law, $t = \frac{1}{4}$ and

$$G((\frac{1}{2}, \frac{5}{8}]) = \frac{35}{36} - \frac{8}{9} = \frac{1}{12} < \frac{7}{36} = \frac{3}{4} - \frac{5}{9} = G_t((\frac{1}{2}, \frac{5}{8}]).$$

A uniform base score is shift-monotone, and M18 gives $D(Q) \geq 0$, so it never meets the condition. □

Proof of M22

Lean: EntryContestPMUWitness.pairA_fosd, EntryContestPMUWitness.pairB_fosd, EntryContestPMUWitness.pairA_step_one, EntryContestPMUWitness.pairA_step_two, EntryContestPMUWitness.pairB_step, EntryContestPMUWitness.fosd_does_not_sign.

Pair one, with $y = 1 - x$, $W_1 = x(1 - x)^2$, $W_2 = x(1 - (1 - x)^2)(1 - x)^2$ and $dG = 2(1 - x)dx$:

$$\int W_1 dF = \frac{1}{12}, \quad \int W_1 dG = \frac{1}{10}, \quad \int W_2 dF = \int_0^1 (y^2 - y^3 - y^4 + y^5) dy = \frac{1}{20}, \quad \int W_2 dG = \frac{11}{210},$$

$$\frac{D(1)}{V} = \frac{1}{10} - \frac{1}{12} = \frac{1}{60}, \quad \frac{D(2)}{V} = \frac{11}{210} - \frac{1}{20} = \frac{1}{420}.$$

Pair two, $W_Q = x^{Q+1} - x^{Q+2}$ and $dF = 2x dx$:

$$\int W_Q dF = \frac{2}{(Q+3)(Q+4)}, \quad \int W_Q dG = \frac{1}{(Q+2)(Q+3)}, \quad \frac{D(Q)}{V} = -\frac{2(Q+2) - (Q+4)}{(Q+2)(Q+3)(Q+4)}.$$

□

Proof of M23

Lean: EntryContestKappa.burden_strictAnti_of_strictConcave, EntryContestBM.burden_strictAnti_of_deriv, EntryContestBM.not_burden_of_convex_stretch, EntryContestKappa.log_burden_strictAnti, EntryContestKappa.crra_strictConcave, EntryContestKappa.crra_burden_strictAnti.

For u strictly concave and $c < a < b$, secant slopes fall as either end moves right, and $a - c < a < b$, $a - c < b - c < b$ give

$$\frac{u(a) - u(a - c)}{c} > \frac{u(b) - u(a - c)}{b - a + c} > \frac{u(b) - u(b - c)}{c},$$

so $\kappa(a) > \kappa(b)$. With u differentiable,

$$\kappa'(w) = u'(w) - u'(w - c) < 0 \quad (w > c).$$

If u is convex on $[a, b]$ with $b - a > c$, then $a < b - c$ and

$$\kappa(a + c) = u(a + c) - u(a) \leq u(b) - u(b - c) = \kappa(b).$$

Log and $x^{1-\gamma}/(1-\gamma)$ have $u'' < 0$ on $(0, \infty)$.

□

Proof of M24

Lean: EntryContestBM.rampU_burden, EntryContestBM.rampU_kappa_diverges, EntryContestBM.rampU_strictConvex, EntryContestBM.rampU_not_concave, EntryContestBM.admissibleParams_open, EntryContestBM.rampU_admissible, EntryContestBM.admissible_width, EntryContestBM.bm_strictly_weaker.

Let ρ rise linearly from 0 at x_0 to 1 at $x_0 + \sigma$, and

$$u(x) = \log x + x + \eta \int_0^x \rho(t) dt, \quad u'(x) = \frac{1}{x} + 1 + \eta \rho(x) > 0.$$

On $[x_0, x_0 + \sigma]$, $u'' = -x^{-2} + \eta/\sigma > 0$ when $\sigma < \eta x_0^2$, so u is strictly convex there and not concave. For $w > c$, $\rho(w) - \rho(w - c) \in [0, 1]$ and vanishes unless $x_0 < w < x_0 + \sigma + c$, where

$$\frac{1}{w-c} - \frac{1}{w} = \frac{c}{w(w-c)} > \frac{c}{(x_0 + \sigma + c)(x_0 + \sigma)} > \eta,$$

so $u'(w) < u'(w - c)$ and M23 applies. $u(x) \rightarrow -\infty$ as $x \downarrow 0$, so M25 gives the divergence. The conditions

$$x_0 > 0, \quad \sigma > 0, \quad \sigma < \eta x_0^2, \quad \eta(x_0 + \sigma + c)(x_0 + \sigma) < c$$

are strict inequalities in continuous functions, so they define an open set, and for $\sigma < c$ an η between σ/x_0^2 and $c/((x_0 + \sigma + c)(x_0 + \sigma))$ exists once x_0 is large, since $\sigma(x_0 + \sigma + c)(x_0 + \sigma)/x_0^2 \rightarrow \sigma < c$.

□

Proof of M25

Lean: EntryContestKappa.kappa_diverges_iff, EntryContestKappa.not_diverges_of_continuousWithinAt_zero, EntryContestKappa.log_kappa_diverges, EntryContestKappa.crra_kappa_diverges, EntryContestKappa.crra_kappa_not_diverges.

$u(w) \rightarrow u(c)$ as $w \downarrow c$, so

$$\kappa(w) = u(w) - u(w - c) \rightarrow \infty \iff u(w - c) \rightarrow -\infty \iff u(x) \rightarrow -\infty (x \downarrow 0).$$

$\log x \rightarrow -\infty$, and

$$\frac{x^{1-\gamma}}{1-\gamma} \rightarrow -\infty \quad (\gamma > 1), \quad \frac{x^{1-\gamma}}{1-\gamma} \rightarrow 0 \quad (\gamma < 1).$$

□

Proof of M26

Lean: EntryContestAnon.profile_mem, EntryContestAnon.profile_not_mem, EntryContestAnon.cdf_profile, EntryContestAnon.gain_anonymous.

Challenger i wins with probability $\int C \prod_{j \neq i} L_j dL_i$, where L_j is F for an entrant and G for an outsider. With $|T| = m$ and $i \notin T$,

$$\frac{\text{gain}_i(T)}{V} = \int C F^m G^{Q-1-m} dF - \int C F^m G^{Q-1-m} dG = \int H_m dF - \int H_m dG = \frac{\Delta(m)}{V}.$$

□

Proof of M27

Lean: EntryContestAnon.winProb_zero_entrant, EntryContestAnon.winProb_zero_outsider, EntryContestAnon.gain_zero, EntryContestAnon.kappa3_values, EntryContestAnon.gains3, EntryContestAnon.anon_witness_zero, EntryContestAnon.gain_tendsto, EntryContestAnon.anon_witness_positive, EntryContestAnon.anon_witness_exists.

Let entrant j 's s_j be the largest of n_j uniform draws, the incumbent's the largest of one, and $\mu = 0$. An outsider scores 0, and among the $1 + \sum_{j \in S} n_j$ draws each is equally likely to be the largest, so

$$\Pr(i \text{ wins}) = \frac{n_i}{1 + \sum_{j \in S} n_j}, \quad \text{gain}_i(T) = \frac{V n_i}{1 + n_i + \sum_{j \in T} n_j}.$$

With $V = 1$, $c = 1$, $u(x) = -1/x$, $w = (\frac{11}{4}, \frac{9}{4}, 2)$ and $n = (1, 2, 3)$,

$$\kappa = (\frac{16}{77}, \frac{16}{45}, \frac{1}{2}), \quad \text{gain}_2(\emptyset) = \frac{3}{4}, \quad \text{gain}_0(\{2\}) = \frac{1}{5}, \quad \text{gain}_1(\{2\}) = \frac{1}{3},$$

$$\text{gain}_0(\{1\}) = \frac{1}{4}, \quad \text{gain}_1(\{0\}) = \frac{1}{2}, \quad \text{gain}_2(\{0, 1\}) = \frac{3}{7},$$

so $\{2\}$ and $\{0, 1\}$ are equilibria. Every inequality is strict and each gain is continuous in μ at 0, so both remain equilibria for small $\mu > 0$.

□

B Evidence and verification

Every row of the model table is a theorem of Lean 4 with Mathlib. The Lean column names the theorems, and the proof of each row in Appendix A names the same ones. The development lives in `lean/mathlib/`, built on the toolchain `v4.24.0-rc1` against Mathlib commit `14871d5`, beside the core file `lean/EntryContest.lean`, which uses no Mathlib. The script `verify.sh` runs three kinds of check on every build, and `VERIFICATION.md` records what they printed.

1. *The axiom audit.* No source file contains `sorry`. Every declared theorem is passed to `#print axioms`. A Mathlib theorem that depends on an axiom other than `propext`, `Classical.choice` and `Quot.sound` fails the build, and a theorem of the core file that depends on an axiom other than `propext` and `Quot.sound` fails it too.
2. *The manuscript checker* (`checks/verify_manuscript.py`). Every model row names audited theorems and has exactly one proof naming the same theorems. Introduction rows point to model rows, and conclusion rows point to model or literature rows. Every literature quote appears verbatim in the cited paper’s record under `lit/`. Plain English is at most 30% of every proof. Each rule also runs on a faulty sample, which the rule must reject.
3. *The literature records.* Each cited paper has a directory under `lit/` holding a reconstruction from the source, the claims our documents make about the paper with page locators, notes on every discrepancy found, and Lean for the mathematical claims. A record checks our reading of a source and does not reprove the source. The records run under `lit/verify_lit.sh`.

What Lean does not carry. The standing assumptions of Appendix D enter the theorems as hypotheses. The rows assume that F and G have no atoms rather than deriving it. The assumption follows from the primitives when r has no atoms and $\mu \neq 0$ (`EntryContestRefute.investorLaw_noAtoms`, `EntryContestRefute.nonInvestorLaw_noAtoms`). A statement about what a cited paper says rests on a quotation with its page, recorded under `lit/`, and not on Lean.

Illustrations. The SymPy and numerical scripts under `checks/` still run in `verify.sh`, as illustrations and as regression tests. No row rests on them. Earlier drafts graded each claim by an evidence tier (Lean, SymPy or numerical) because several analytic steps entered the Lean development only as hypotheses. Those steps are now proved in Lean, and the tiers are retired.

C Terminology

Every English term that carries load in a row or a proof is bound here to a symbol built from the primitives of Appendix D. Each entry names an object already defined, so the table is a translation key and not a second set of definitions. Indices follow Appendix A, with the richest challenger at index 0.

Term	Symbol	Meaning
<i>Players and actions</i>		
challenger	$i \in \{0, \dots, Q-1\}$	A player of given wealth who chooses whether to compete.
incumbent	C	The additional player, who competes in every case and appears only through the score CDF C .
enter	$e_i = 1$	Pay the fee c and add $(1 - \mu)s_i$ to the score.
entrant, outsider	$e_i = 1, e_i = 0$	A challenger who enters, and one who does not.
entrant set	$S = \{i : e_i = 1\}$	The challengers who enter.
prize	$V > 0$	Fixed and awarded by rank, outside u .
<i>Wealth and cost</i>		
wealth	w_i	Challenger i ’s resource. A challenger with $w_i \leq c$ does not enter.
wealth profile	$(w_0, \dots, w_{Q-1})$	The realised wealths, common knowledge, sorted with $w_0 \geq w_1 \geq \dots$.
entry cost	$\kappa(w) = u(w) - u(w - c)$	The burden of the fee in utility units.
fee	c	The nominal cost of entry, the same for every challenger.
burden-monotonicity	κ strictly decreasing on (c, ∞)	The assumption on u (M23), weaker than strict concavity (M24).
<i>Scores and gains</i>		
score	$\sigma_i = \mu r_i + (1 - \mu)s_i e_i$	The prize goes to the highest score.
score CDFs	F, G	The laws of an entrant’s score and an outsider’s, with $F \leq G$ (M2).
dominance gap	$\phi = G - F \geq 0$	What entry buys.
best rival score	$H_m = CF^m G^{Q-1-m}$	The CDF of the best score facing a challenger when m others enter.
gain	$\Delta(m)$	The value of entering when m others enter, with $\Delta(m+1) \leq \Delta(m)$ (M4).

Term	Symbol	Meaning
anonymity	Δ depends on m alone	The gain depends on how many others enter and not on who they are (M26, M27).
first entrant's gain	$\Delta_Q(0), D(Q)$	The gain with no other entrant among Q challengers, and its step in Q (M16 to M22).
<i>Equilibrium</i>		
entry condition	$\kappa_j \leq \Delta(j)$	Index j enters when its cost is at most its gain.
entry count	k^*	The number of indices before the first one whose entry condition fails (M5).
count invariance	$ S = k^*$	Every pure-strategy equilibrium has k^* entrants (M6).
identity multiplicity	$S \neq S', S = S' = k^*$	Different entrant sets can be equilibria at the same count (Appendix D).
assortative set	$\{0, \dots, k^* - 1\}$	The k^* richest challengers (M7, M8).
marginal entrant	index $k^* - 1$	The poorest entrant of the assortative set.
first outsider	index k^*	The richest challenger outside the assortative set.
<i>Comparative statics</i>		
change of profile	$w \mapsto w'$	Any map between sorted wealth profiles. No result assumes it keeps the mean unless it says so.
mean-preserving spread	$\sum_j w'_j = \sum_j w_j, w'$ majorizing w	The case of economic interest, and not a hypothesis of M12.
displacement	$d_j = w'_j - w_j$	The rank-by-rank difference of the two sorted profiles.
pivot-spread	T with $T(w) \geq w$ for $w \geq x_0$ and $T(w) \leq w$ for $w \leq x_0$	A non-decreasing map that spreads wealth on both sides of the pivot x_0 (M13).
ordering in dispersion	$z - y$ non-decreasing	For quantile functions y and z , the second law is more dispersed [Lewis and Thompson, 1981]. With a crossing point, the transport map between the two laws is a pivot-spread.
margin condition	$d_{k^*-1} \geq 0$, or $d_{k^*} \leq 0$	The hypothesis of the rise branch, read at the marginal entrant, and of the fall branch, read at the first outsider (M12).
band	$d_{k^*-1} < 0 < d_{k^*}$	The margins on which the margin condition is silent (M14).
<i>Aggregates</i>		
total entry cost	$\sum_{i \in S} \kappa(w_i)$	In utility units. Its resource counterpart $ S c = k^*c$ is the same in every equilibrium.
sum of payoffs	$\sum_i U_i$	Unweighted, so an interpersonal comparison of the index u (M8).
selection	a rule picking one equilibrium	Not a refinement. Nothing claims that players coordinate on it.

Three pairs are kept apart because earlier drafts conflated them. The entry cost in utility, κ , varies with wealth, while the fee in resources, c , does not and is what society gives up. Count invariance holds, while identity multiplicity is real, and every comparative static concerns the count for that reason. A change of profile is what the results apply to, while the margin condition is what they assume.

D Primitives and scope

The contest. One prize of value $V > 0$ is fixed and exogenous, and it is awarded by rank rather than sold. Q challengers and one incumbent compete. Each challenger chooses $e_i \in \{0, 1\}$, and entry costs the fee c . The draws r_i and $s_i \geq 0$ are independent across players and independent of the wealth profile, from arbitrary laws. The score is $\sigma_i = \mu r_i + (1 - \mu)s_i e_i$, the highest score takes the prize, and the game is static and played once. By M1 the gain is proportional to V , so V enters only as a scale. Cole et al. [1992] read status as “a ranking device that determines how well he or she fares in the nonmarket sector”, which is the sense of a prize awarded by rank.

Payoffs. With u the utility of wealth,

$$U_i = u(w_i - c e_i) + V \mathbf{1}\{i \text{ wins}\}.$$

The fee is paid inside u , so its burden is felt in utility and varies with wealth. The prize enters outside u and linearly, so rich and poor value it alike. Inequality therefore acts through one channel, the cost of entry. This is the privilege contest of Schroyen and Treich [2016], and not their rent-seeking contest, in which a money prize would sit inside u .

Utility. u is strictly increasing and burden-monotone, so $\kappa(w) = u(w) - u(w - c)$ is strictly decreasing on (c, ∞) . Strict concavity implies it (M23), and it is strictly weaker (M24). A challenger whose wealth is at or below c does not enter. The strictness result M15 also uses $\kappa(w) \rightarrow \infty$ as $w \downarrow c$, which holds exactly when $u(x) \rightarrow -\infty$ as $x \downarrow 0$ (M25).

Score laws. F is the law of $\mu r + (1 - \mu)s$ and G the law of μr , and both are taken to have no atoms (Appendix B). Earlier drafts also assumed a common lower endpoint for F and G and that ϕ vanishes at both ends of the support. No row uses either, and both are dropped.

Information. The realised wealth profile is common knowledge, and entry is simultaneous. Under this assumption the entry condition is written on the sorted profile, and k^* is a count shared by every pure-strategy equilibrium (M6). It does not pin who enters. With $\kappa = (2, 5, 8)$ and $\Delta = (12, 10, 1)$, the sets $\{0, 1\}$, $\{0, 2\}$ and $\{1, 2\}$ are all equilibria, the last without the richest challenger (EntryContestEq.three_equilibria). Identities are pinned when $\Delta(k^* - 1) < \kappa_{k^*}$ (M7). Under private entry costs the count would be binomial and the entrant set random, as in Moreno and Wooders [2011], where every buyer compares her own cost with one threshold. Here the threshold depends on the index, $\kappa_j \leq \Delta(j)$, and it is flat only when Δ is constant, which M4 rules out whenever $\int \phi^2 dK_m > 0$.

The anonymity boundary. Independence of the draws from wealth makes F and G , and so Δ , depend on the count of entrants alone (M26). Economies in which wealth buys ability are outside every row. The exclusion is shown and not only asserted. When ability falls with wealth, equilibria of different sizes coexist (M27), and count invariance fails. The question of how many enter and the question of which enter then cease to be separable, and every comparative static of the model concerns the count. When ability rises with wealth, count invariance held in every economy sampled by checks/verify_anonymity.py. That direction is not claimed.

E Refuted conjectures

R1, a collapse threshold. An earlier draft conjectured a finite $\bar{Q}$ beyond which $k^* = 0$. The conjecture is false. With r uniform on $[0, 1]$, $s = 0$ with probability p and $s = 1$ otherwise, $0 < \mu \leq \frac{1}{2}$ and an investing incumbent, for every $Q \geq 1$

$$\Delta_Q(0) = V \left(\frac{1-p^2}{2} - \frac{p(1-p)}{Q+1} \right) \geq V \frac{1-p}{2}, \quad \Delta_Q(0) \rightarrow V \frac{1-p^2}{2},$$

so a challenger whose cost is at most $V(1-p)/2$ enters for every Q (EntryContestRefute.bern_Delta, EntryContestRefute.bern_Delta_lower, EntryContestRefute.bern_Delta_limit). When an outsider's score never exceeds x_G , $\Delta(m) \geq V \left(\int_{x \geq x_G} CF^m dF - \frac{1}{Q-m} \right)$ for $m < Q$, and the first term does not depend on Q (EntryContestRefute.r1_lower_bound). An earlier draft gave the limit as $\Pr(X > \mu)$ times the chance of beating the incumbent. That formula is withdrawn, since in the family above $\Pr(X > \mu) = 1 - p$ (EntryContestRefute.bern_investor_above) while the limit is $V(1-p)(1+p)/2$.

R2, fewer entrants as Q grows. A referee conjectured that for $\mu > 0$ a larger Q shrinks the gain, so fewer enter. The conjecture is not general. In the family of R1 with $0 < p < 1$, $\Delta_Q(0)$ rises strictly at every Q (EntryContestRefute.bern_Delta_strictMono), so a challenger whose cost lies between $\Delta_Q(0)$ and $\Delta_{Q+1}(0)$ stays out with Q challengers and enters with $Q+1$. With r uniform the gain never falls in Q (M18). The conjectured direction occurs for other primitives (M20), and the sign of the effect is set by the law of r (M16 to M22).

F Responses to the referee reports

Asymmetric k -of- Q equilibria. Both referees found that the earlier version compared aggregate participation regimes instead of checking unilateral deviations. The model now derives the asymmetric equilibrium in which k^* of the Q challengers enter, checks deviations index by index, and shows the count is the same in every equilibrium (M5 to M8).

Prize and affordability. The second referee objected that the earlier version used one quantity both as inequality and as the prize, which made the inequality result close to tautological, and that a larger prize should raise every participant’s incentive. The prize is now fixed and separate from the wealth distribution (Appendix D). A larger prize raises every gain and does not lower the entry count, and a larger fee does not raise it (M11). The inequality result is the sign rule of M12 to M15, in which V does not move.

The mobility ceiling. Both referees found the old ceiling an artefact of uniform supports and of a limit in Q under which the equilibrium does not exist. The saturation bound replaces it. An entrant at index m with cost κ has $(m+1)\kappa \leq V$, so the marginal entrant gives $k^* \leq V/\kappa_{k^*-1}$ for every Q , with no limit taken (M10).

G Coverage of the referee comments

The two reports on the earlier submission (BEJEAP, rejected 31 August 2026) are recorded comment by comment in `HANDOVER_referee_reports_20260831.md`. Each comment is marked *answered* when a model row or an appendix settles it, *partly answered* when part of it remains, *open* when nothing here settles it, and *moot* when it concerned text or an object that the rebuilt model no longer has. Comments that only praised a change are omitted.

Comment	Substance	Status	Where
<i>Reviewer 1</i>			
R1.1	Aggregate regimes were compared in place of unilateral deviations.	answered	Deviations are checked index by index. The count k^* is the same in every pure-strategy equilibrium, and the assortative set is one (M5 to M8).
R1.2	The reach into education policy is limited by a zero-sum, fixed-slot contest.	answered	The rebuilt model makes no policy claim. The prize is fixed and awarded by rank, and the scope is stated with it (Appendix D).
R1.3	Results were asserted beyond uniform distributions without proof.	answered	Every row holds for general laws of r and s unless it names a family or a class, and the rows that do say so (M18 to M22, M27).
R1.4	The contribution is not located against contests with entry costs, positional education and status competition.	partly answered	The literature table (L1 to L13) places each result. The novelty reading of the remaining papers is pending, and the conclusions table waits for it.
R1.4	The mobility ceiling reflects the uniform specification.	answered	The ceiling is replaced by a bound for every Q and every law, with no limit taken (M10).
R1.a2	“Stable Nash equilibrium” is undefined.	answered	Only pure-strategy Nash equilibrium is used (M6).
R1.a3	Full withdrawal should be a third regime.	answered	The count k^* ranges over $0, \dots, Q$, so no entry, partial entry and full entry are all cases of one result (M5, M6).
R1.a4	The ceiling is used inconsistently across regimes.	answered	One bound covers every count (M10).
R1.a5	Prize effects and affordability effects are conflated.	answered	The prize sits outside u and the fee inside it (Appendix D). The prize and the fee move the count separately (M11).
R1.a6	The notation $\Phi(Q)$ is inconsistent.	moot	The rebuilt model has no Φ .
R1.a7	Comparative statics in μ were misstated.	open	No comparative static in μ is claimed.
R1.a8	The paper is verbose.	answered	Results sit in tables, and every proof is at most 30% prose, checked by script (Appendix B).
<i>Reviewer 2</i>			
R2.1	Asymmetric equilibria in which only some challengers enter.	answered	The equilibrium is the count k^* of entrants, and every equilibrium has it (M5, M6).
R2.1	More enter as the prize rises and as the fee falls.	answered	M11.
R2.1	More enter as social capital matters more (μ falls).	open	Not claimed.
R2.1	At $\mu = 0$ the number entering does not depend on Q .	answered	At $\mu = 0$ the gain is $V/(m+2)$ for every Q (M9), so with the same costs the count is the same.
R2.1	At $\mu > 0$ a larger Q means fewer enter.	answered	Not in general. The first entrant’s gain can rise in Q , never falls with a uniform base score, and the sign depends on the law of r (M16 to M22, conjecture R2 in Appendix E).
R2.1	An equilibrium in which the incumbent abstains and a challenger enters.	open	The incumbent’s entry is exogenous here. Endogenising it is item O1 of <code>TOD0.md</code> .

Comment	Substance	Status	Where
R2.2	The effect of inequality on status expenditure is unclear, and the prize is the inequality.	answered	The prize is fixed and apart from the wealth distribution. A spread changes the count, and so the resource expenditure k^*c , with a sign set by the marginal entrant against the pivot (M12 to M15).
R2.2	Endowment inequality should reduce status expenditure, as in the literature.	answered	It does when the marginal entrant is at or below the pivot, and the reverse holds when it is above (M13).
R2.3	Policy is discussed without an objective.	partly answered	No policy claim is made. Among equilibria of one size the assortative set has the least total utility cost, and the resource cost k^*c is the same in all of them (M6, M8). No welfare section is attempted.
R2.4	The ceiling is an artefact, inconsistent with equilibrium, and set in an artificial comparative static.	answered	M10 bounds k^* for every Q with no limit taken.
R2.5	A static one-shot model would be simpler.	answered	The game is static and played once (Appendix D).
R2.5	Give the incumbent a lower cost instead of relying on concavity.	answered	M5 to M8 hold for any ordered costs, whatever their source, and concavity is replaced by burden-monotonicity (M23, M24).
R2.5	Financial constraints are not crucial.	answered	No budget constraint is used beyond a challenger at or below the fee not entering (Appendix D).
R2.5	Consider a reduced-form contest success function.	partly answered	The score is kept, and every result is stated in the laws F , G and C of the scores.
R2.5	Why is $\mu < \frac{1}{2}$ needed?	answered	No row assumes it. The family of conjecture R1 uses $\mu \leq \frac{1}{2}$ only to be explicit.
R2.6	Results should be precise propositions with proofs.	answered	Each model row states hypotheses and a conclusion and has a proof checked in Lean (Appendix A).
R2.6	Errors in the text of the earlier manuscript (saving, p_1 , marginal utility cost, income ratios, free riding, the Sutton Trust passage).	moot	The text is retired. One point carries over. A uniform rise in every wealth lowers every cost, so the count does not fall (M12).

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